Symolore AI Training Course schedule

Introduction to Generative AI for Developers

Course Number: AI-120WA
Duration: 40 Hours

Overview

This Introduction Generative AI (GenAI) training teaches developers how to build intelligent, scalable applications using GenAI and large language models (LLMs).

Prerequisites

- Practical experience in Python (at least 6 months) would be beneficial:
    - Data Structures, Functions, Control Structures
    - Exception Handling, File I/O, async, concurrency (recommended)
- Practical experience with these Python libraries: Pandas, NumPy, and scikit-learn would be beneficial
    - Understanding of Machine Learning concepts - regression, clustering, classification
    - ML Algorithms: Gradient Descent, Linear Regression

Materials

All Generative AI training students receive comprehensive courseware.

Software Needed on Each Student PC

All attendees must have a modern web browser and an Internet connection.

Objectives

- Understand the architecture and capabilities of Large Language Models (LLMs)
- Integrate LLMs into applications using popular frameworks like LangChain with OpenAI, GoogleAIStudio etc.

Outline

Python for Generative AI (4 Hours)

- Python basics for AI
- Introduction to Jupyter Notebooks
- Libraries: NumPy, pandas, matplotlib

Basics of ML/DL/AI (6 Hours)

- Introduction to Machine Learning (ML), Deep Learning (DL), and AI
- Supervised vs. Unsupervised Learning
- Neural Networks and key ML/DL algorithms
- Vectors and Embeddings
- Attention Mechanism and Transformers

Copyright ©2003-2024 Accelebrate, LLC. Some outlines may contain content from our courseware partners; such content is protected by these partners' copyrights. All trademarks are owned by their respective owners.

- Libraries: (scikit-learn, PyTorch, HuggingFace Transformers)

Introduction to Generative AI and LLMs (4 Hours)

- Overview of Generative AI
- Basics of LLMs and their architecture
- Popular LLM architectures (e.g., GPT, BERT, T5)
- Quantization
- Roles and Conversation Threading
- Popular LLMs, APIs, and Libraries - Generative AI Tech Stack

LangChain Fundamentals (6 Hours)

- Introduction and Setting up LangChain
- LCEL (LangChain Expression Language)
- Basic operations and Chain
- Observability (LangSmith, LangFuse)
- Integrating with APIs (OpenAI, Gemini, HuggingFace etc)

Prompt Engineering (2 Hours)

- Fundamentals of prompt engineering
- Prompting Techniques (ZeroShot, FewShot, CoT, ReAct etc)

Projects with Advanced LangChain (8 Hours)

- VectorStore and Retriever
- Tools, Agents and Memory
- RAG (Retrieval-Augmented Generation)
- Customizing LangChain modules
- Developing chatbots and conversational agents
- Creating question-answering systems using RAG
- Building task-oriented applications

Popular GenAI Frameworks & Tools (2 Hours)

- Overview of popular frameworks (LangChain, LlamaIndex, CrewAI)
- Introduction to relevant tools and libraries

Training LLMs (4 Hours)

- Basics of training LLMs
- Fine-tuning and transfer learning
- Training techniques and strategies
- Practical training examples

Copyright ©2003-2024 Accelebrate, LLC. Some outlines may contain content from our courseware partners; such content is protected by these partners' copyrights. All trademarks are owned by their respective owners.

Packaging and Deploying in Production (2 Hours)

- LLMOps
- Model deployment strategies
- Scaling and maintaining models

Future Direction and Conclusion (2 Hours)

- Research and Future trends in Generative AI and LLMs
- Ethical considerations and responsible AI
- Course review and next steps

Copyright ©2003-2024 Accelebrate, LLC. Some outlines may contain content from our courseware partners; such content is protected by these partners' copyrights. All trademarks are owned by their respective owners.

Copyright ©2003-2024 Accelebrate, LLC. Some outlines may contain content from our courseware partners; such content is protected by these partners' copyrights. All trademarks are owned by their respective owners.